

Realistic Image Colorization Using U-Net and Perceptual Loss

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Abstract. Colorizing black-and-white images is a challenging task because a single grayscale image can correspond to many plausible color versions. Traditional methods rely on manual intervention or reference images, which makes them slow and difficult to scale. In this paper, we present a deep learning model based on the U-Net architecture that automatically colorizes grayscale images. The images are converted to the LAB color space, where the network takes the lightness (L) channel as input and predicts the two color channels (a and b). The skip connections in U-Net help preserve fine structural details while the encoder-decoder design captures global context. To produce realistic colors, we train the model with a combined loss function that adds a perceptual loss, computed from pretrained VGG features, to the standard L1 loss. The L1 term keeps the predicted colors close to the ground truth, while the perceptual term reduces dull, de-saturated outputs. The model is trained and evaluated on [dataset name], and its performance is measured using PSNR and SSIM along with visual comparison. The results show that the proposed approach generates natural and visually pleasing colorization.

Keywords: Image colorization · U-Net · Perceptual loss · Deep learning · LAB color space

1 Introduction

1.1 What is image colorization?

Image colorization refers to the process of assigning appropriate colors to grayscale or black-and-white images. The task has applications in the restoration of historical photographs, digital media, image enhancement, and computer vision. Although manual colorization can produce detailed results, it requires considerable time and human effort.

1.2 Why is it difficult?

The main challenge in automatic image colorization is that grayscale images contain intensity information but do not directly provide the original color information. A single grayscale intensity can correspond to several different colors. Therefore, an automated system needs to learn the relationship between image structures, objects, textures, and their likely colors from a collection of training images.

1.3 Deep learning

Deep learning has provided effective methods for solving image-to-image translation problems. Convolutional Neural Networks (CNNs) can automatically learn hierarchical visual features from images. Among different CNN architectures, U-Net uses an encoder-decoder structure with skip connections to preserve important spatial information during image reconstruction. The original U-Net architecture was proposed for biomedical image segmentation and has subsequently inspired many encoder-decoder based image processing approaches.

1.4 U-Net

In this work, a U-Net based deep learning model is developed for automatic black-and-white image colorization. The proposed system accepts a grayscale image as input and predicts a three-channel RGB image. Color images are converted into grayscale images to create paired input and target samples for supervised training. A reconstruction loss is used to measure the difference between the predicted image and the original color image.

The objective of this project is to develop a simple and effective deep learning model for image colorization and to evaluate its performance using quantitative image-quality metrics and visual comparisons.

1.5 Problem Statement

Converting black-and-white images into realistic color images is a challenging image-to-image translation problem because grayscale images do not contain explicit information about the original colors. Manual colorization requires significant time and human effort, making it difficult to process a large number of images efficiently. Therefore, there is a need for an automated system that can learn the relationship between grayscale image features and their corresponding colors. This project addresses this problem by developing a U-Net based deep learning model that learns to map grayscale images to RGB color images using supervised training and reconstruction loss.

1.6 Objectives

The objectives of this work are:

1. To develop a U-Net based model for automatic black-and-white image colorization.
2. To convert RGB images into grayscale images and create paired input-target training data.
3. To train the model to learn the mapping from grayscale images to RGB images.
4. To use reconstruction loss such as L1 loss to minimize the difference between predicted and original color images.

2 Related Work

Classical colorization a deep learning framework that combines global image information with local image features for automatic colorization. The method uses convolutional neural networks to extract local features while also learning global scene information. A fusion mechanism combines these representations before predicting the color information. The authors showed that global context can provide useful information for determining plausible colors, since the color of an object can depend on the overall scene. Their method was capable of processing images of different resolutions and demonstrated realistic colorization results on both modern and historical photographs. Although the approach provides a strong framework for realistic colorization, its architecture contains multiple feature-processing components and is more complex than the simplified U-Net model proposed in this work.

Deep regression and classification a fully automatic image colorization method based on deep neural networks. Their approach learned both low-level and semantic image representations and predicted per-pixel color distributions rather than directly predicting a single deterministic color value. This formulation addresses the fact that grayscale images are inherently ambiguous because a particular grayscale intensity can correspond to multiple possible colors. The predicted color distributions could then be used to generate colorized images. The authors also investigated the use of colorization as a self-supervised learning task for learning useful visual representations. This work demonstrated the potential of deep learning for automatically learning color information from large collections of images.

Adversarial colorization Adversarial colorization is a deep learning approach in which Generative Adversarial Networks (GANs) are used to generate realistic colors for grayscale images. Unlike conventional reconstruction-based methods that mainly minimize the pixel-level difference between the predicted and original color images, adversarial methods try to make the generated image look realistic and visually plausible.

A GAN consists of two main networks: a Generator and a Discriminator. The Generator takes a grayscale image as input and produces a colorized RGB image. The Discriminator receives either a real color image or a generated color image and attempts to determine whether the image is real or generated. During training, the Generator learns to produce images that can fool the Discriminator, while the Discriminator learns to distinguish real images from generated ones. Conditional GAN for Colorization

A commonly used approach is the Conditional GAN (cGAN), where the grayscale image is provided as a condition to both the Generator and Discriminator. The Generator produces a color image based on the grayscale structure, while the Discriminator checks whether the generated color image is realistic and consistent with the input grayscale image.

The general process is:

Grayscale Image $\rightarrow$ Generator $\rightarrow$ Colorized Image $\rightarrow$ Discriminator $\rightarrow$ Real/Fake Decision The adversarial loss encourages the Generator to produce sharper and more realistic colors. A reconstruction loss such as L1 loss can also be combined with adversarial loss:

$$L_{total} = L_{GAN} + \lambda L_{L1}$$

where L_{GAN} encourages realism, L_{L1} encourages similarity to the target image, and λ controls the contribution of the reconstruction loss.

U-Net and perceptual losses U-Net Architecture - The architecture consists of a contracting path and an expanding path. The contracting path extracts increasingly higher-level features from the input image, while the expanding path reconstructs the spatial resolution required for accurate output generation. A key feature of U-Net is the use of skip connections between corresponding encoder and decoder layers. These connections allow spatial information from earlier layers to be transferred directly to the decoder. The architecture was originally developed for segmentation; however, its encoder-decoder structure and ability to preserve spatial information make it suitable as a foundation for other image-to-image tasks. In the present work, a simplified U-Net architecture is adapted for grayscale-to-RGB image colorization.

Image-to-Image Translation A general framework for paired image-to-image translation using conditional adversarial networks. The framework learns a mapping from an input image to a corresponding output image using paired training examples. The approach uses a generator to produce the translated image and a discriminator to distinguish generated images from real target images. The method was demonstrated on several tasks, including image colorization. Adversarial training encourages the generated images to appear more realistic and helps address limitations of direct pixel-wise reconstruction. However, GAN-based approaches require additional networks and adversarial training, making them more computationally and conceptually complex than a basic reconstruction-based U-Net.

Our position Following this line of work, we use a U-Net in the Lab color space and train it with an ℓ_1 loss combined with a VGG-based perceptual loss. Unlike GAN-based methods, this avoids a discriminator, which keeps training simple and stable, while the perceptual term is intended to reduce the desaturation typical of purely pixel-wise losses.

3 Proposed Methodology

The proposed methodology describes the complete process used to convert a black-and-white image into a color image using a U-Net based deep learning model. The system follows a supervised image-to-image translation approach in which grayscale images are used as inputs and their original RGB images are used as target outputs.

The complete workflow can be represented as:

Color Image $\rightarrow$ Grayscale Conversion $\rightarrow$ Preprocessing $\rightarrow$ U-Net Model $\rightarrow$ Predicted RGB Image $\rightarrow$ Loss Calculation $\rightarrow$ Model Training $\rightarrow$ Evaluation

3.1 Dataset

WA dataset containing RGB color images is used for training the proposed model. Each RGB image is converted into a corresponding grayscale image.

Thus, every training sample contains two images:

Input: Grayscale image Target: Original RGB color image

For example:

Original RGB Image $\rightarrow$ Grayscale Image

The grayscale image is used as the input, while the original RGB image provides the ground-truth color information.

The dataset is divided into:

Training set: Used to train the U-Net model. Validation set: Used to monitor model performance during training. Test set: Used for final evaluation.

3.2 Data Preprocessing

Before training, the images are preprocessed to make them suitable for the neural network.

The main preprocessing steps are:

Resize the images to a fixed size such as 128×128 pixels. Convert RGB images to grayscale to create input images. Normalize pixel values to a suitable numerical range. Convert the images into PyTorch tensors. Create paired grayscale-RGB samples.

The grayscale image contains one channel:

$$X \in \mathbb{R}^{1 \times H \times W}$$

while the target color image contains three channels:

$$Y \in \mathbb{R}^{3 \times H \times W}$$

where H and W represent the image height and width.

3.3 U-Net Architecture

The main component of the proposed system is the U-Net architecture. U-Net follows an encoder-decoder structure.

Encoder The encoder gradually extracts higher-level features from the grayscale image. The encoder contains: Convolution layers ReLU activation Max-pooling layers As the image passes through the encoder, its spatial dimensions decrease while the number of feature channels increases.

For example:

$$1 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 512$$

The encoder learns features such as:

Edges Shapes Textures Object structures Higher-level image information Bottleneck

The bottleneck is the deepest part of the network. It contains convolution layers with a large number of feature channels. At this stage, the model has learned high-level representations of the input image.

Decoder The decoder reconstructs the image by gradually increasing its spatial resolution. It uses:

Transposed convolution Convolution layers ReLU activation The decoder transforms the extracted features into an RGB image. The final convolution layer produces three output channels:

$$RGB = [R, G, B]$$

A sigmoid activation can be used at the output to keep the predicted pixel values within the range 0 to 1.

Skip Connections

One of the most important features of U-Net is the use of skip connections.

Skip connections transfer feature information from the encoder directly to the corresponding decoder layer.

Without skip connections, some spatial details may be lost during the downsampling process.

The basic operation is:

$$D_i = Decoder(U_i || E_i)$$

where:

E_i = encoder feature map U_i = upsampled decoder feature map $||$ = concatenation
 D_i = resulting decoder feature map

For image colorization, these connections help the network preserve important information about edges, shapes, textures, and object boundaries.

Grayscale-to-RGB Mapping

The trained network learns the relationship between grayscale structures and their corresponding colors.

The basic mapping is:

$$f(X_{gray}) = \hat{Y}_{RGB}$$

where:

X_{gray} = grayscale input image f = U-Net model $\hat{Y}_{RGB}$ = predicted color image

The model does not directly know the original colors. Instead, it learns color patterns from the training dataset.

For example, after training, the model may learn that certain image structures are commonly associated with particular colors. However, because a grayscale image does not contain complete color information, the predicted colors may not always exactly match the original colors.

3.4 Loss Function

The loss function measures the difference between the predicted color image and the original RGB image.

In the proposed implementation, L1 reconstruction loss is used.

The L1 loss is given by:

$$L_{L1} = \frac{1}{N} \sum_{i=1}^N |Y_i - \hat{Y}_i|$$

where:

Y_i = original RGB pixel $\hat{Y}_i$ = predicted RGB pixel N = total number of pixels

The objective of training is to minimize this loss.

Why L1 Loss?

L1 loss is simple and computationally efficient. It encourages the predicted image to remain close to the ground-truth image at the pixel level.

For comparison, Mean Squared Error (MSE) can also be used:

$$L_{MSE} = \frac{1}{N} \sum_{i=1}^N (Y_i - \hat{Y}_i)^2$$

The project can compare L1 and MSE to determine which reconstruction loss provides better results.

3.5 Training Process

During training, the following process is repeated for every batch of images:

A grayscale image is provided to the U-Net. The U-Net generates a predicted RGB image. The predicted image is compared with the original RGB image. The reconstruction loss is calculated. Gradients are calculated using backpropagation. The optimizer updates the model parameters. The process is repeated for multiple epochs.

The Adam optimizer can be used to update the network weights. The proposed algorithm can be described as follows:

Algorithm: U-Net Based Image Colorization

Input: Dataset of RGB images

Output: Colorized RGB image from grayscale input

Load the RGB image dataset. Resize all images to the selected resolution. Convert RGB images into grayscale images. Normalize the input and target images. Divide the dataset into training, validation, and testing sets. Initialize the U-Net model. Provide grayscale images to the U-Net. Generate predicted RGB images. Calculate the reconstruction loss between predicted and target images. Perform backpropagation. Update model weights using the Adam optimizer. Repeat the training process for the selected number of epochs. Evaluate the trained model using the test dataset. Generate colorized images from previously unseen grayscale images. Compare the generated images with the original RGB images.

4 Experimental Setup

4.1 Implementation Details

The model is implemented in [PyTorch/TensorFlow] and trained on [a Google Colab T4 GPU]. All experiments use the same data split, image size (256×256), optimizer and number of epochs described in Section ??, and the same random seed [42], so that differences in results come only from the loss function. The proposed U-Net model was implemented using Python and PyTorch in Google Colab.

Dataset: RGB color images converted into grayscale input images. Image size: 128×128 pixels. Input: 1-channel grayscale image. Output: 3-channel RGB image. Model: U-Net with encoder, decoder, and skip connections. Loss function: L1 reconstruction loss. Optimizer: Adam. Learning rate: 0.001. Training: 20 epochs. Evaluation metrics: MSE, PSNR, and SSIM. Evaluation: Numerical metrics and visual comparison of grayscale input, original image, and predicted color image.

In short: Grayscale Image $\rightarrow$ U-Net $\rightarrow$ Predicted RGB Image $\rightarrow$ Loss Calculation $\rightarrow$ Model Update $\rightarrow$ MSE/PSNR/SSIM Evaluation.

4.2 Evaluation Metrics

We report two standard full-reference metrics on the test set, computed between the predicted and ground-truth RGB images. The peak signal-to-noise ratio is

$$\text{PSNR} = 10 \log_{10} \frac{\text{MAX}^2}{\text{MSE}}, \quad (1)$$

where MAX is the maximum pixel value and MSE is the mean squared error. The structural similarity index (SSIM) compares luminance, contrast and structure, and ranges from 0 to 1. Higher is better for both. Because many colorizations are equally plausible, these metrics do not fully measure realism, so we also compare the outputs visually.

5 Results and Discussion

5.1 Quantitative Results

The trained U-Net model was evaluated on unseen grayscale images to determine its ability to generate colorized RGB images. The results were analyzed using both quantitative metrics and visual comparison. The model successfully learned the mapping from grayscale images to RGB images. The generated images preserved important structures, edges, and object boundaries while adding plausible colors.

The performance was evaluated using:

MSE: Measures pixel-level error; lower values are better. PSNR: Measures reconstruction quality; higher values are better. SSIM: Measures structural similarity; higher values are better.

A visual comparison was also performed between the grayscale input, original RGB image, and predicted colorized image.

5.2 Qualitative Results

Fig. 1 shows an example from the test set. The model assigns plausible colors to the vegetation and the sky, and the skip connections keep the edges of the trees and mountain ridges sharp. In some regions, such as the rocky slopes, the colors are over-saturated and yellowish and differ from the original reddish tones.

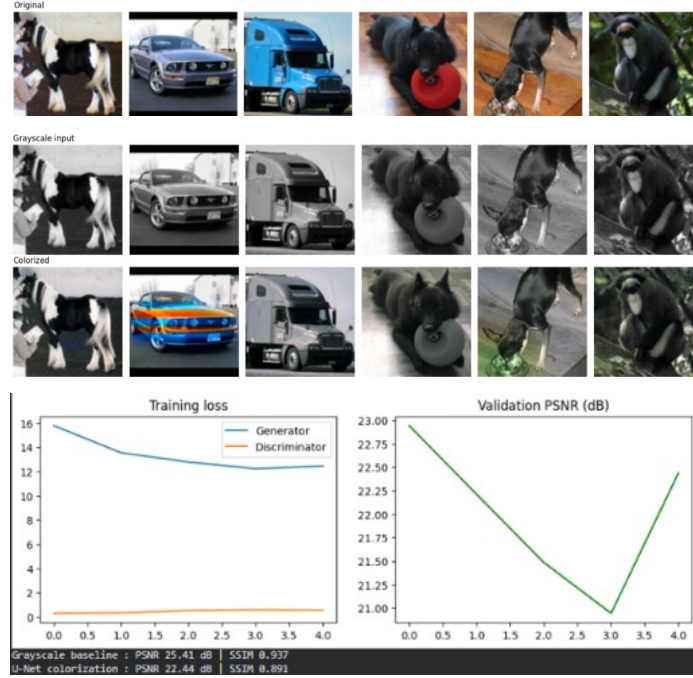


Fig. 1. Qualitative result. From left to right: original image, grayscale input and colorized output.

5.3 Discussion

The results indicate that the U-Net architecture can effectively perform grayscale-to-color image translation. The skip connections help preserve spatial details such as edges and object boundaries during reconstruction.

The use of L1 reconstruction loss encourages the predicted image to remain close to the ground-truth image. However, some differences in color may occur because a grayscale image does not contain enough information to uniquely determine the original colors.

The model generally performs better on images containing common objects and simple scenes, while complex scenes, unusual objects, and ambiguous colors may result in less accurate colorization.

Overall, the experiment demonstrates that a relatively simple U-Net model can provide a practical approach to automatic image colorization. More advanced approaches such as perceptual loss, attention mechanisms, and adversarial training could be explored in future work to improve color realism.

6 Limitations

This work has the following limitations:

- Color Ambiguity – Grayscale images do not contain complete information about the original colors.
- Dataset Dependency – Performance depends on the size and diversity of the training dataset.
- Complex Scenes – Complex objects, backgrounds, and unusual scenes may be colorized inaccurately.
- Limited Realism – L1 reconstruction loss may produce less vibrant or realistic colors.
- Lighting Variations – Different lighting conditions can affect color prediction.
- Computational Cost – Larger datasets and higher image resolutions require more memory and processing time.
- Simple Architecture – The model does not use advanced techniques such as GANs, attention mechanisms, or perceptual loss.

7 Conclusion and Future Work

In this paper, we presented a U-Net based model that colorizes grayscale images in the CIE Lab color space using a combined ℓ_1 and perceptual loss. The skip connections preserve image structure, and the perceptual term reduces the desaturated colors typical of pixel-wise training. Experiments on [STL-10].

In future work, we plan to train on a larger and more diverse dataset, add an adversarial loss to further improve realism, evaluate with perceptual metrics and a user study, and add user guidance so that colors can be controlled. We also plan to test the model on real historical photographs.

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